

Homework 1 – Clustering & Disease Phenotyping

Instructions on submission:

- You must upload a .zip file with your name in the title that includes:
 - **Written answers:** a .pdf file with your written answers
 - **Code:** A Jupyter notebook (.ipynb) with your code. Please use the template notebook provided.
 - **Plots:** this can be separate from the notebook (i.e. generated plots in figs folder) or saved in the notebook with the outputs displayed
 - Please **do not** include the dataset in your submission

Other instructions:

- Code must run. Missing code will result in no credit for relevant written answers.
- Label all plots (titles, axes, legends). Unlabeled plots will result in point deductions.

Note: you may use any libraries but using Scanpy + Scikit-Learn is the most straightforward for this file format. You may reference the following tutorials to get familiar with the tools available in Scanpy and Scikit-learn:

- [Scanpy tutorial 1](#)
- [Scanpy tutorial 2](#)
- [Scikit-learn tutorial](#)

Overview

You are working as a computational medicine researcher studying aging in mice. Your PI gives you access to the [Tabula Muris Senis](#) dataset (single-cell RNA-seq across 20 mouse organs) and asks whether you can identify tissue-specific phenotypes and cell-type clusters. The project goal is to understand whether aging alters clustering of single-cell transcriptomes across different tissues.

You will use dimensionality reduction + clustering to group cells, then identify marker genes for clusters.

Questions

Conceptual (20 pts) - written in PDF

1. (3 points) Define a *clinical phenotype*. Why is unsupervised clustering useful for discovering disease subtypes?
2. (3 points) Compare k-means vs hierarchical clustering. What assumptions do they make about cluster shape/structure?
3. (4 points) How can dimensionality reduction (PCA) help in analyzing high-dimensional scRNA-seq data?
4. (5 points) In disease phenotyping using cluster analysis, why is it important to identify clinically recognizable clusters? Put differently: what is/are the problems if the identified clusters are not clinically meaningful?
5. (5 points) Partitioning Around Medoids (PAM) with Gower's distance is designed for clustering mixed data types. Why did K-means with Euclidean distance perform better on the clinical data (with mixed data types) in Lecture 5? Similarly, why did K-means with Euclidean distance outperform K-means with Manhattan distance on the same dataset?

Coding, Analysis, and Interpretation (80 pts)

1. (5 points) Data preprocessing
 - a. Select three tissues (Limb_Muscle, Pancreas, Spleen) with ~5,000 cells total (just download the provided data).
 - b. Normalize counts, log-transform, and filter out low-quality cells (simply run the provided code).
2. (20 points) Dimensionality reduction
 - a. Apply PCA (with 50 components) and plot the first 2 PCs, coloring points by *tissue label*.
 - b. *Interpretation (written in PDF)*: Do the tissues separate in PCA space? Explain your observation
3. (20 points) K-means clustering
 - a. Run k-means clustering on PCA-reduced data from the previous step
 - b. Use the silhouette score or elbow method to decide K
 - c. Provide a PCA plot with clusters overlaid
 - d. *Interpretation (written in PDF)*: How well do clusters align with tissue labels?
4. (20 points) Alternative clustering method
 - a. Apply either hierarchical clustering or a graph-based clustering algorithm (e.g., Louvain algorithm) as an alternative clustering approach.
 - b. Compare performance to k-means using internal metrics (e.g.,

silhouette).

- c. *Interpretation (written in PDF)*: How does your alternative clustering method compare to k-means clustering?
5. (15 points) Marker gene identification
- a. For each cluster, identify top 5 marker genes using differential expression analysis (fold change and adjusted p-value of a statistical test). Rank genes based on fold change, with a significance threshold of adjusted p-value < 0.05 . (Most of the code is provided)
 - b. *Interpretation (written in PDF)*: Are the identified markers similar for different tissues? Explain briefly why they are or are not similar.